Лабораторная работа №2

Тема: Настройка виртуальной локальной сети (VLAN)

1) Для заданной на схеме schema-lab3 сети, состоящей из управляемых коммутаторов и персональных компьютеров настроить на коммутаторах логическую топологию используя протокол IEEE 802.1Q, для передачи пакетов VLAN333 между коммутаторами использовать Native VLAN

Из проекта Menzhulin-lab3-template была скопирована сеть (рис. 1).

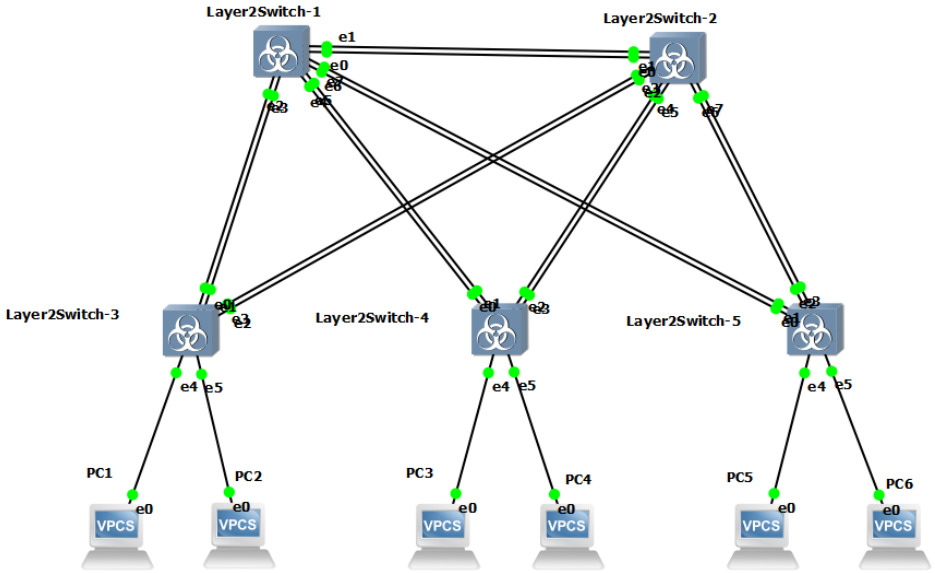

Рис. 1. Топология сети.

Настройка vlan’ов между коммутаторами.

На всех коммутаторах надо выполнить инструкции ниже для интерфейсов между ними.

```
Switch>enable
Switch#conf t
Switch(config)#interface range GigabitEthernet 1/0-3
Switch(config-if-range)#switchport trunk encapsulation dot1q
Switch(config-if-range)#switchport mode trunk
Switch(config-if-range)#switchport trunk native vlan 333
Switch(config-if-range)#switchport trunk allowed vlan 20,333
Switch(config-if-range)#no shutdown
Switch(config-if-range)#exit
Switch(config)#exit
```

На коммутаторах Layer2Switch-3 – 5 надо выполнить инструкции ниже для интерфейсов для VPC.

```
Switch(config)#interface GigabitEthernet 1/0
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#no shu
Switch(config-if)#no shutdown
Switch(config-if)#exit

Switch(config)#interface GigabitEthernet 1/1
Switch(config-if)#switchport mode acces
Switch(config-if)#switchport access vlan 333
Switch(config-if)#no shutdown
Switch(config-if)#exit
```

Проверка настроек

```
Switch#show interfaces trunk

Port        Mode             Encapsulation  Status        Native
vlan
Gi0/0       on               802.1q         trunking      333
Gi0/1       on               802.1q         trunking      333
Gi0/2       on               802.1q         trunking      333
Gi0/3       on               802.1q         trunking      333
Gi1/0       on               802.1q         trunking      333
Gi1/1       on               802.1q         trunking      333
Gi1/2       on               802.1q         trunking      333
Gi1/3       on               802.1q         trunking      333

Port        Vlans allowed on trunk
Gi0/0       20,333
Gi0/1       20,333
Gi0/2       20,333
Gi0/3       20,333
Gi1/0       20,333
Gi1/1       20,333
Gi1/2       20,333
Gi1/3       20,333

Switch#show interfaces status

Port      Name       Status       Vlan       Duplex  Speed Type
Gi0/0               connected    trunk        auto   auto unknown
Gi0/1               connected    trunk        auto   auto unknown
Gi0/2               connected    trunk        auto   auto unknown
Gi0/3               connected    trunk        auto   auto unknown
Gi1/0               connected    20           auto   auto unknown
Gi1/1               connected    333          auto   auto unknown
```

2) Проверить доступность персональных компьютеров, находящихся в одинаковых VLAN и недоступность находящихся в различных, результаты задокументировать

Настройка VPC

Hostname	IP-address	Subnet Mask
PC1	192.168.10.1	255.255.255.0
PC2	192.168.20.1	255.255.255.0
PC3	192.168.10.2	255.255.255.0
PC4	192.168.20.2	255.255.255.0
PC5	192.168.10.3	255.255.255.0
PC6	192.168.20.3	255.255.255.0

На PC1

```
PC1> ip 192.168.10.1 255.255.255.0
Checking for duplicate address...
PC1 : 192.168.10.1 255.255.255.0

PC1> save
Saving startup configuration to startup.vpc
.  done

PC1> show ip

NAME        : PC1[1]
IP/MASK     : 192.168.10.1/24
GATEWAY     : 255.255.255.0
DNS         :
MAC         : 00:50:79:66:68:00
LPORT       : 20240
RHOST:PORT  : 127.0.0.1:20241
MTU         : 1500
```

Проверка связи

На PC1

```
PC1> ping 192.168.10.2

84 bytes from 192.168.10.2 icmp_seq=1 ttl=64 time=14.384 ms
84 bytes from 192.168.10.2 icmp_seq=2 ttl=64 time=10.346 ms
84 bytes from 192.168.10.2 icmp_seq=3 ttl=64 time=4.557 ms
84 bytes from 192.168.10.2 icmp_seq=4 ttl=64 time=7.975 ms
84 bytes from 192.168.10.2 icmp_seq=5 ttl=64 time=6.167 ms

PC1> ping 192.168.10.3

84 bytes from 192.168.10.3 icmp_seq=1 ttl=64 time=9.090 ms
84 bytes from 192.168.10.3 icmp_seq=2 ttl=64 time=3.717 ms
84 bytes from 192.168.10.3 icmp_seq=3 ttl=64 time=6.052 ms
84 bytes from 192.168.10.3 icmp_seq=4 ttl=64 time=1.728 ms
84 bytes from 192.168.10.3 icmp_seq=5 ttl=64 time=12.704 ms

PC1> ping 192.168.20.3

host (255.255.255.0) not reachable

PC1> ping 192.168.20.1

host (255.255.255.0) not reachable

PC1> ping 192.168.20.2
```

```
host (255.255.255.0) not reachable
```

Связь внутри VLAN20 есть, а с VLAN333 нету.

На PC4

```
PC4> ping 192.168.10.1

host (255.255.255.0) not reachable

PC4> ping 192.168.10.2

host (255.255.255.0) not reachable

PC4> ping 192.168.10.3

host (255.255.255.0) not reachable

PC4> ping 192.168.20.3

84 bytes from 192.168.20.3 icmp_seq=1 ttl=64 time=7.701 ms
84 bytes from 192.168.20.3 icmp_seq=2 ttl=64 time=6.928 ms
84 bytes from 192.168.20.3 icmp_seq=3 ttl=64 time=5.397 ms
84 bytes from 192.168.20.3 icmp_seq=4 ttl=64 time=9.124 ms
84 bytes from 192.168.20.3 icmp_seq=5 ttl=64 time=1.355 ms

PC4> ping 192.168.20.1

84 bytes from 192.168.20.1 icmp_seq=1 ttl=64 time=11.863 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=64 time=7.575 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=64 time=5.815 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=64 time=7.508 ms
84 bytes from 192.168.20.1 icmp_seq=5 ttl=64 time=6.286 ms
```

Связь внутри VLAN333 есть, а с VLAN20 нету.

3) Перехватить в WireShark пакеты с тегами и без тегов (nb!), результаты задокументировать

Для захвата пакетов с тегом использовался фильтр vlan, а для захвата без тега !vlan. Доп. фильтрация шла по протоколу icmp, через который идёт ping и arp для того чтобы увидеть, что обратной связи из другого VLAN нет.

Ниже на рисунках 2 и 3 представлены захваченные пакеты.

vlan && (icmp || arp)

No.	Time	Source	Destination	Protocol	Lengtl	Info
17	12.666778	Private_66:68:00	Broadcast	ARP	68	Who has 192.168.10.2? Tell 192.168.10.1
18	12.669707	Private_66:68:02	Private_66:68:00	ARP	68	192.168.10.2 is at 00:50:79:66:68:02
19	12.672554	192.168.10.1	192.168.10.2	ICMP	102	Echo (ping) request id=0x312c, seq=1/256
20	12.677539	192.168.10.2	192.168.10.1	ICMP	102	Echo (ping) reply id=0x312c, seq=1/256
21	13.678448	192.168.10.1	192.168.10.2	ICMP	102	Echo (ping) request id=0x322c, seq=2/512
22	13.681703	192.168.10.2	192.168.10.1	ICMP	102	Echo (ping) reply id=0x322c, seq=2/512
26	14.682794	192.168.10.1	192.168.10.2	ICMP	102	Echo (ping) request id=0x332c, seq=3/768
27	14.685276	192.168.10.2	192.168.10.1	ICMP	102	Echo (ping) reply id=0x332c, seq=3/768
29	15.687995	192.168.10.1	192.168.10.2	ICMP	102	Echo (ping) request id=0x342c, seq=4/102
30	15.691095	192.168.10.2	192.168.10.1	ICMP	102	Echo (ping) reply id=0x342c, seq=4/102
33	16.692282	192.168.10.1	192.168.10.2	ICMP	102	Echo (ping) request id=0x352c, seq=5/128
34	16.701254	192.168.10.2	192.168.10.1	ICMP	102	Echo (ping) reply id=0x352c, seq=5/128
89	57.226684	Private_66:68:00	Broadcast	ARP	68	Who has 192.168.10.3? Tell 192.168.10.1
90	57.228580	Private_66:68:04	Private_66:68:00	ARP	68	192.168.10.3 is at 00:50:79:66:68:04
91	57.229874	192.168.10.1	192.168.10.3	ICMP	102	Echo (ping) request id=0x5e2c, seq=1/256
92	57.240886	192.168.10.3	192.168.10.1	ICMP	102	Echo (ping) reply id=0x5e2c, seq=1/256
95	58.242575	192.168.10.1	192.168.10.3	ICMP	102	Echo (ping) request id=0x5f2c, seq=2/512
96	58.244020	192.168.10.3	192.168.10.1	ICMP	102	Echo (ping) reply id=0x5f2c, seq=2/512
97	59.252237	192.168.10.1	192.168.10.3	ICMP	102	Echo (ping) request id=0x602c, seq=3/768
98	59.257459	192.168.10.3	192.168.10.1	ICMP	102	Echo (ping) reply id=0x602c, seq=3/768
101	60.266399	192.168.10.1	192.168.10.3	ICMP	102	Echo (ping) request id=0x612c, seq=4/102
102	60.274599	192.168.10.3	192.168.10.1	ICMP	102	Echo (ping) reply id=0x612c, seq=4/102
103	61.275859	192.168.10.1	192.168.10.3	ICMP	102	Echo (ping) request id=0x622c, seq=5/128
104	61.282318	192.168.10.3	192.168.10.1	ICMP	102	Echo (ping) reply id=0x622c, seq=5/128
115	68.618582	Private_66:68:00	Broadcast	ARP	68	Who has 255.255.255.0? Tell 192.168.10.1
116	69.618960	Private_66:68:00	Broadcast	ARP	68	Who has 255.255.255.0? Tell 192.168.10.1
119	70.619970	Private_66:68:00	Broadcast	ARP	68	Who has 255.255.255.0? Tell 192.168.10.1
128	76.358878	Private_66:68:00	Broadcast	ARP	68	Who has 255.255.255.0? Tell 192.168.10.1
129	77.359356	Private_66:68:00	Broadcast	ARP	68	Who has 255.255.255.0? Tell 192.168.10.1
133	78.359874	Private_66:68:00	Broadcast	ARP	68	Who has 255.255.255.0? Tell 192.168.10.1
389	248.906800	Private_66:68:00	Broadcast	ARP	68	Who has 255.255.255.0? Tell 192.168.10.1
390	249.906894	Private_66:68:00	Broadcast	ARP	68	Who has 255.255.255.0? Tell 192.168.10.1
393	250.907237	Private_66:68:00	Broadcast	ARP	68	Who has 255.255.255.0? Tell 192.168.10.1
402	257.138691	Private_66:68:00	Broadcast	ARP	68	Who has 192.168.10.3? Tell 192.168.10.1
403	257.140574	Private_66:68:04	Private_66:68:00	ARP	68	192.168.10.3 is at 00:50:79:66:68:04
404	257.145890	192.168.10.1	192.168.10.3	ICMP	102	Echo (ping) request id=0x262d, seq=1/256
405	257.150464	192.168.10.3	192.168.10.1	ICMP	102	Echo (ping) reply id=0x262d, seq=1/256
408	258.151417	192.168.10.1	192.168.10.3	ICMP	102	Echo (ping) request id=0x272d, seq=2/512
409	258.153299	192.168.10.3	192.168.10.1	ICMP	102	Echo (ping) reply id=0x272d, seq=2/512
411	259.158835	192.168.10.1	192.168.10.3	ICMP	102	Echo (ping) request id=0x282d, seq=3/768
412	259.166137	192.168.10.3	192.168.10.1	ICMP	102	Echo (ping) reply id=0x282d, seq=3/768
415	260.167493	192.168.10.1	192.168.10.3	ICMP	102	Echo (ping) request id=0x292d, seq=4/102
416	260.174031	192.168.10.3	192.168.10.1	ICMP	102	Echo (ping) reply id=0x292d, seq=4/102
417	261.174979	192.168.10.1	192.168.10.3	ICMP	102	Echo (ping) request id=0x2a2d, seq=5/128
418	261.181881	192.168.10.3	192.168.10.1	ICMP	102	Echo (ping) reply id=0x2a2d, seq=5/128

Рис. 2. Пакеты с тегами vlan

!vlan && (icmp || arp)

No.	Time	Source	Destination	Protocol	Lengtl	Info
209	137.270740	Private_66:68:01	Broadcast	ARP	64	Who has 192.168.20.2? Tell 192.168.20.1
210	137.283733	Private_66:68:03	Private_66:68:01	ARP	64	192.168.20.2 is at 00:50:79:66:68:03
211	137.285315	192.168.20.1	192.168.20.2	ICMP	98	Echo (ping) request id=0xae2c, seq=1/256
212	137.287668	192.168.20.2	192.168.20.1	ICMP	98	Echo (ping) reply id=0xae2c, seq=1/256
217	138.288584	192.168.20.1	192.168.20.2	ICMP	98	Echo (ping) request id=0xaf2c, seq=2/512
218	138.298713	192.168.20.2	192.168.20.1	ICMP	98	Echo (ping) reply id=0xaf2c, seq=2/512
219	139.300441	192.168.20.1	192.168.20.2	ICMP	98	Echo (ping) request id=0xb02c, seq=3/768
220	139.302596	192.168.20.2	192.168.20.1	ICMP	98	Echo (ping) reply id=0xb02c, seq=3/768
223	140.303770	192.168.20.1	192.168.20.2	ICMP	98	Echo (ping) request id=0xb12c, seq=4/102
224	140.304997	192.168.20.2	192.168.20.1	ICMP	98	Echo (ping) reply id=0xb12c, seq=4/102
225	141.305892	192.168.20.1	192.168.20.2	ICMP	98	Echo (ping) request id=0xb22c, seq=5/128
226	141.308117	192.168.20.2	192.168.20.1	ICMP	98	Echo (ping) reply id=0xb22c, seq=5/128
249	157.762680	192.168.20.1	192.168.20.2	ICMP	98	Echo (ping) request id=0xc22c, seq=1/256
250	157.764057	192.168.20.2	192.168.20.1	ICMP	98	Echo (ping) reply id=0xc22c, seq=1/256
253	158.770018	192.168.20.1	192.168.20.2	ICMP	98	Echo (ping) request id=0xc32c, seq=2/512
254	158.779594	192.168.20.2	192.168.20.1	ICMP	98	Echo (ping) reply id=0xc32c, seq=2/512
255	159.780832	192.168.20.1	192.168.20.2	ICMP	98	Echo (ping) request id=0xc42c, seq=3/768
256	159.789656	192.168.20.2	192.168.20.1	ICMP	98	Echo (ping) reply id=0xc42c, seq=3/768
259	160.791744	192.168.20.1	192.168.20.2	ICMP	98	Echo (ping) request id=0xc52c, seq=4/102
260	160.794051	192.168.20.2	192.168.20.1	ICMP	98	Echo (ping) reply id=0xc52c, seq=4/102
261	161.796022	192.168.20.1	192.168.20.2	ICMP	98	Echo (ping) request id=0xc62c, seq=5/128
262	161.797265	192.168.20.2	192.168.20.1	ICMP	98	Echo (ping) reply id=0xc62c, seq=5/128
881	613.595299	Private_66:68:03	Broadcast	ARP	64	Who has 255.255.255.0? Tell 192.168.20.2
885	614.600172	Private_66:68:03	Broadcast	ARP	64	Who has 255.255.255.0? Tell 192.168.20.2
886	615.595701	Private_66:68:03	Broadcast	ARP	64	Who has 255.255.255.0? Tell 192.168.20.2

Рис. 3. Пакеты без тегов vlan.

Как видно из изображений, ответ на arp запрос происходит только внутри одной VLAN сети. При этом vlan протокол использует только VLAN-20, т.к. VLAN333 используется native vlan.

Пакет VLAN-20

У всех сообщений в этой vlan появляется новое поле внутри протокола Ethernet

```
Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst:
Private_66:68:02 (00:50:79:66:68:02)
    Destination: Private_66:68:02 (00:50:79:66:68:02)
    Source: Private_66:68:00 (00:50:79:66:68:00)
    Type: 802.1Q Virtual LAN (0x8100)
    [Stream index: 4]

802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 20
    000. .... .... .... = Priority: Best Effort (default) (0)
    ...0 .... .... .... = DEI: Ineligible
    .... 0000 0001 0100 = ID: 20
    Type: IPv4 (0x0800)
```

В то время, как у сообщений в сети VLAN-333 их нет.

```
Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst:
Private_66:68:03 (00:50:79:66:68:03)
    Destination: Private_66:68:03 (00:50:79:66:68:03)
        .... ..0. .... .... .... .... = LG bit: Globally unique
address (factory default)
```

```
        .... ...0 .... .... .... .... = IG bit: Individual address
(unicast)
    Source: Private_66:68:01 (00:50:79:66:68:01)
        .... ..0. .... .... .... .... = LG bit: Globally unique
address (factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address
(unicast)
    Type: IPv4 (0x0800)
    [Stream index: 9]
```